Chapter 3
Numerical Methods for ODE IVPs

Ordinary differential equations are divided into two classes: *initial value problems* (IVPs), where initial conditions for the solution are specified at some time t_0, and *boundary value problems* (BVPs), where boundary conditions for the solution are specified at the left and right boundaries. This chapter will discuss numerical methods for ODE initial value problems, and the next chapter numerical methods for ODE boundary value problems.

An ODE initial value problem could equivalently be formulated for a solution depending on x, where initial conditions for the solution are specified at some spatial point x_0.

To solve the ODE initial value problem

$$\frac{du}{dt} = f(u), \quad u(t = 0) = u_0 \tag{3.1}$$

we use a one-step method and approximate

$$\frac{du}{dt} \approx \frac{u_{n+1} - u_n}{\Delta t} = \tilde{f}(u_n, u_{n+1}, \Delta t) \tag{3.2}$$

where $\tilde{f}(u_n, u_{n+1}, \Delta t)$ is an approximation to the right-hand side of the initial value problem (3.1). Unfolding the time derivative, we obtain the standard form of the numerical method:

$$u_{n+1} = u_n + \Delta t\, \tilde{f}(u_n, u_{n+1}, \Delta t). \tag{3.3}$$

Different numerical methods are defined by different approximations $\tilde{f}$. If $\tilde{f} = f(u_n)$, we have the forward Euler method, which is *explicit* since the right-hand side of (3.3) only involves the old time level value u_n. If $\tilde{f}$ involves u_{n+1}, the method is called *implicit*. Backward Euler, where $\tilde{f} = f(u_{n+1})$, and TR, where

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Table 3.1 Numerical methods for ODE initial value problems. Stability is for $du/dt = au, a < 0$

Method	Type	Order	Stability
forward Euler	Explicit	First	$\Delta t \leq 2/\lvert a\rvert$
backward Euler	Implicit	First	L-stable
TR	Implicit	Second	A-stable
MMR	Implicit	Second	A-stable
predictor-corrector	Explicit	Second	$\Delta t \leq 2/\lvert a\rvert$
RK2	Explicit	Second	$\Delta t \leq 2/\lvert a\rvert$
RK4	Explicit	Fourth	$\Delta t \leq 2.79/\lvert a\rvert$
TRBDF2	Implicit	Second	L-stable

$\tilde{f} = \frac{1}{2}(f(u_n) + f(u_{n+1}))$, are examples of implicit methods. Table 3.1 lists the numerical methods we will discuss in this chapter.

We will develop time-dependent numerical methods for deterministic dynamical systems and analyze the geometry of the phase space of solution trajectories, including chaotic solution behavior and strange attractors in the Shaw oscillator and Lorenz equations. At the end of the chapter, we prove the Equivalence Theorem for ODE initial value and boundary value problems.

See Iserles' *A First Course in the Numerical Analysis of Differential Equations* [22] and Butcher's *Numerical Methods for Ordinary Differential Equations* [5] for general introductions to solving ordinary differential equations.

3.1 Forward and Backward Euler Methods

The *forward Euler* method for $du/dt = f(u)$ is

$$u_{n+1} = u_n + \Delta t\, f(u_n). \tag{3.4}$$

Forward Euler is an explicit method and is first-order accurate and conditionally stable.

For the linear case $du/dt = au$, the exact solution is $u(t) = u_0 e^{at}$. The growth factor for forward Euler is defined by

$$u_{n+1} = (1 + a\Delta t)u_n = Gu_n. \tag{3.5}$$

The exact growth factor $H = e^{a\Delta t}$. Note that $G = e^{a\Delta t}$ to first order in Δt (*consistency*). For n timesteps,

$$u_n = G^n u_0 = (1 + a\Delta t)^n u_0 \to e^{na\Delta t} u_0 = e^{at_n} u_0 = u(t_n) \tag{3.6}$$

as $n \to \infty$ and $\Delta t \to 0$ with $n\Delta t = t_n$ held fixed. We have proved u_n *converges* to the exact solution $u(t_n)$ in this limit. (Henceforth we will prove that a numerical

method is consistent and stable; then convergence follows from the Equivalence Theorem.)

To analyze *stability*, we consider the model problem

$$\frac{du}{dt} = au, \quad a < 0 \quad (\textit{model problem}) \tag{3.7}$$

because then the exact solution $u(t) = e^{-|a|\Delta t}u_0$ decays. The computed solution $u_n = G^n u_0$ must not grow for $a < 0$, so stability requires $|G| \leq 1$. For forward Euler, we have $-1 \leq 1 + a\Delta t = 1 - |a|\Delta t \leq 1$ or $\Delta t \leq 2/|a|$ for stability. The neutrally stable case in which $|G| = 1$ requires caution, since the numerical solution may be totally inaccurate (see Problem 3.8).

To find the order of accuracy for the full nonlinear problem (3.1), we calculate the global truncation error τ:

$$u(t + \Delta t) = u + \Delta t\, f(u) + \Delta t\, \tau \tag{3.8}$$

where t is a shorthand for t_n and u is a shorthand for $u(t_n)$. Taylor expanding (and using $f(u) = du/dt \equiv u'$), we get

$$u + \Delta t\, u' + \frac{1}{2}\Delta t^2\, u'' + \cdots = u + \Delta t\, u' + \Delta t\, \tau \tag{3.9}$$

$$\tau = \frac{1}{2}\Delta t\, u'' + O(\Delta t^2) \tag{3.10}$$

which shows forward Euler is first-order accurate.

The *backward Euler* method for $du/dt = f(u)$ is

$$u_{n+1} = u_n + \Delta t\, f(u_{n+1}). \tag{3.11}$$

Backward Euler is an implicit method, first-order accurate, and A-stable and L-stable (see Problems 3.1 and 3.2).

For linear systems of ODEs $u' = Au$, the model problem (3.7) is replaced by $u' = \lambda u$, where λ is an eigenvalue of A. For nonlinear systems of ODEs $u' = f(u)$, we linearize and examine the eigenvalues λ of the Jacobian $J = \partial f/\partial u$ in (2.23) $\delta u' = J\delta u$ when $\delta u' = \lambda \delta u$. Note that λ will in general be complex even for real matrices A or J. Thus, we can plot the stability region where the complex growth factor $G(z)$, $z = \lambda\Delta t$, has modulus $|G(z)| \leq 1$ in the complex plane. The stability regions for forward and backward Euler, TR, and TRBDF2 are plotted in Fig. 3.1. A-stability corresponds now to a stability region that includes the entire left half-plane $\Re\{z\} \leq 0$. Thus backward Euler is A-stable, but forward Euler is only conditionally stable.

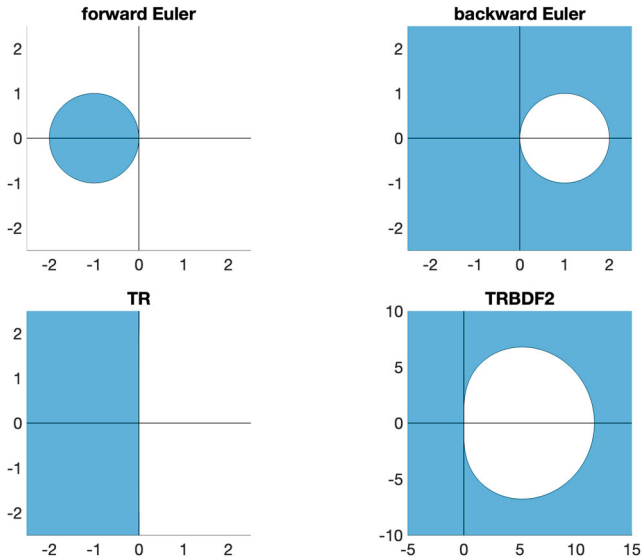

Fig. 3.1 Stability regions (*shaded*) in the complex plane showing only conditional stability for the forward Euler method, and A-stability for the backward Euler, trapezoidal rule, and TRBDF2 ($\gamma = 2 - \sqrt{2}$) methods. Backward Euler and TRBDF2 are also L-stable

3.2 Trapezoidal Rule Method

The *trapezoidal rule (TR)* method for $du/dt = f(u)$ is

$$u_{n+1} = u_n + \frac{\Delta t}{2}(f(u_n) + f(u_{n+1})). \tag{3.12}$$

TR is implicit, second-order accurate, and A-stable but not L-stable.

Stability: The growth factor for TR for $du/dt = au$ is

$$u_{n+1} = \frac{1 + \frac{1}{2}a\Delta t}{1 - \frac{1}{2}a\Delta t}u_n = Gu_n. \tag{3.13}$$

For the model problem ($a < 0$),

$$-1 \le \frac{1 - \frac{1}{2}|a|\Delta t}{1 + \frac{1}{2}|a|\Delta t} = \frac{1 - \Delta}{1 + \Delta} \le 1, \quad \Delta = \frac{1}{2}|a|\Delta t > 0 \tag{3.14}$$

is always true, independent of Δt. TR is not L-stable since

$$\lim_{\Delta t \to \infty} G = -1. \tag{3.15}$$

Consistency: The LTE (= $\Delta t\ \tau$) is given by

$$u(t+\Delta t) = u + \frac{\Delta t}{2}(f(u) + f(u(t+\Delta t))) + \Delta t\ \tau. \tag{3.16}$$

Taylor expanding (and using $f(u) = u'$), we get

$$u+\Delta t\ u'+\frac{1}{2}\Delta t^2\ u''+\frac{1}{6}\Delta t^3\ u'''+\cdots = u+\Delta t\ u'+\frac{1}{2}\Delta t^2\ u''+\frac{1}{4}\Delta t^3\ u'''+\cdots+\Delta t\ \tau \tag{3.17}$$

$$\tau = -\frac{1}{12}\Delta t^2\ u''' + O(\Delta t^3) \tag{3.18}$$

which shows TR is second-order accurate.

The stability region for TR is shown in Fig. 3.1, demonstrating graphically that TR is A-stable.

3.3 Predictor-Corrector Methods

Predictor-corrector methods are frequently used in fluid dynamical computations.

The simplest predictor-corrector method is the second-order (explicit) Adams-Bashforth-Moulton method. For $du/dt = f(t, u)$, the predictor step is a forward Euler extrapolation:

$$\tilde{u}_{n+1} = u_n + \Delta t\ f(t_n, u_n) \tag{3.19}$$

which is followed by a second-order accurate corrector step:

$$u_{n+1} = u_n + \frac{\Delta t}{2}\left(f(t_n, u_n) + f(t_{n+1}, \tilde{u}_{n+1})\right). \tag{3.20}$$

Note the similarities and differences with the TR (3.12), RK2 (3.23), and MMR (Problem 3.3) methods. Problem 3.4 demonstrates that this predictor-corrector method is second-order accurate and conditionally stable.

A fourth-order (explicit) Adams-Bashforth-Moulton method can be obtained by fitting the predictor and corrector steps by cubic polynomials in f. The predictor step fits a cubic polynomial through the four most recent points in the derivative picture, extrapolates it, and integrates under the cubic (see Problem 3.5):

$$\begin{aligned}\tilde{u}_{n+1} = u_n + \frac{\Delta t}{24}\,&(55 f(t_n, u_n) - 59 f(t_{n-1}, u_{n-1}) \\ &+37 f(t_{n-2}, u_{n-2}) - 9 f(t_{n-3}, u_{n-3}))\,.\end{aligned} \tag{3.21}$$

The corrector step also fits a cubic polynomial through the four most recent points in the derivative picture and integrates under the cubic:

$$u_{n+1} = u_n + \frac{\Delta t}{24}\,(9 f(t_{n+1}, \tilde{u}_{n+1})$$
$$+19 f(t_n, u_n) - 5 f(t_{n-1}, u_{n-1}) + f(t_{n-2}, u_{n-2}))\,. \tag{3.22}$$

However, this step does not involve extrapolation, since we now know $f(t_{n+1}, \tilde{u}_{n+1})$.

For both predictor-corrector methods, the timestep is adjusted dynamically based on the methods of Sect. 3.6, approximating $||e_l|| \approx ||u_{n+1} - \tilde{u}_{n+1}||$.

3.4 Runge-Kutta Methods

The *RK2* method for $du/dt = f(u)$ is

$$u_{n+1} = u_n + \Delta t\, f\left(u_n + \frac{1}{2}\Delta t\, f(u_n)\right). \tag{3.23}$$

RK2 is explicit, second-order accurate, and conditionally stable: $\Delta t \le 2/|a|$ for $f(u) = au$, $a < 0$ (see Problem 3.6).

If $f = f(t, u)$ depends on u and explicitly also on t (nonautonomous form), then the RK2 method is

$$u_{n+1} = u_n + \Delta t\, f\left(t_n + \frac{1}{2}\Delta t, u_n + \frac{1}{2}\Delta t\, f(t_n, u_n)\right). \tag{3.24}$$

Note that a nonautonomous ODE $du/dt = f(t, u)$ can always be transformed into a pair of autonomous ODEs by setting $v = t$:

$$\begin{bmatrix} du/dt \\ dv/dt \end{bmatrix} = \begin{bmatrix} f(v, u) \\ 1 \end{bmatrix}, \quad \begin{bmatrix} u(t_0) \\ v(t_0) \end{bmatrix} = \begin{bmatrix} u_0 \\ t_0 \end{bmatrix}. \tag{3.25}$$

The *RK4* method for $du/dt = f(u)$ is

$$u_{n+1} = u_n + \frac{\Delta t}{3}(K_1 + 2K_2 + 2K_3 + K_4) \tag{3.26}$$

with

$$K_1 = \frac{1}{2} f(u_n), \quad K_2 = \frac{1}{2} f(u_n + \Delta t\, K_1),$$

$$K_3 = \frac{1}{2} f(u_n + \Delta t\, K_2), \quad K_4 = \frac{1}{2} f(u_n + 2\Delta t\, K_3). \tag{3.27}$$

RK4 is explicit, fourth-order accurate, and conditionally stable (see Problem 3.7).

3.5 Harmonic Oscillator

As a practical application, we will solve the dynamical equation for the motion of a damped harmonic oscillator:

$$\frac{d^2 y}{dt^2} + 2p\frac{dy}{dt} + \omega_0^2 y = 0 \tag{3.28}$$

where for frictional damping $p > 0$ and where ω_0 is the natural angular frequency of the oscillator without friction, with initial conditions $y(0) = y_0$, $y'(0) = v_0$.

To convert this second-order ODE to a first-order system of ODEs, set $u(t) = y(t)$ and $v(t) = y'(t)$. Then

$$\frac{du}{dt} = v \tag{3.29}$$

$$\frac{dv}{dt} = -2pv - \omega_0^2 u \tag{3.30}$$

with initial conditions $u(0) = y_0$, $v(0) = v_0$. Defining $w(t)$ as the column vector $(u(t), v(t))$, we can write the two first-order ODEs in vector form as

$$\frac{dw}{dt} = \frac{d}{dt}\begin{bmatrix} u \\ v \end{bmatrix} = f(w) = \begin{bmatrix} v \\ -2pv - \omega_0^2 u \end{bmatrix} \tag{3.31}$$

with initial conditions $w(0) = w_0 = (u_0, v_0)$. Since the damped harmonic oscillator ODE is linear,

$$f(w) = \begin{bmatrix} 0 & 1 \\ -\omega_0^2 & -2p \end{bmatrix} w = Aw. \tag{3.32}$$

Modern methods for time-dependent problems always employ a variable Δt adjusted dynamically based on an estimate of the local error. Our model program **rk2dyn.m** implements this idea. The computed and exact solutions for the damped harmonic oscillator are displayed in Fig. 3.2.

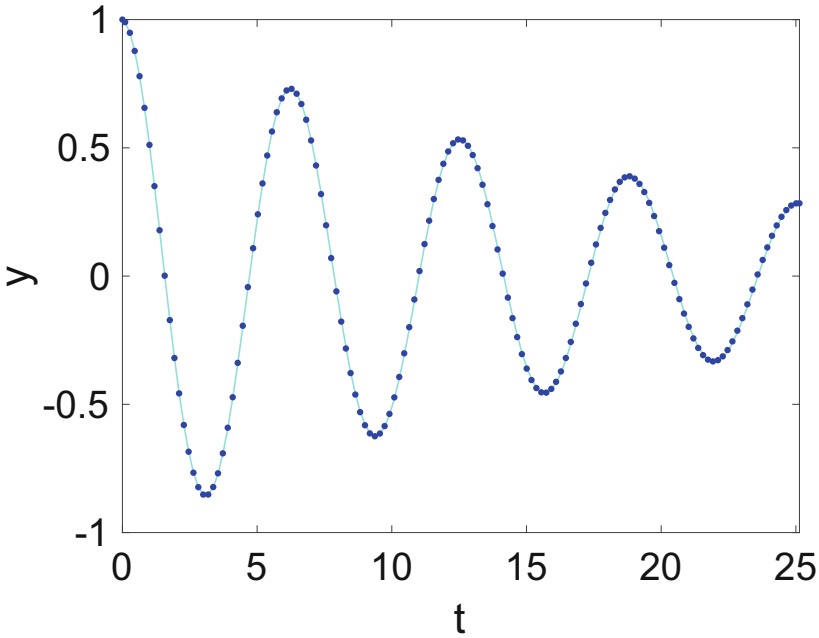

Fig. 3.2 Harmonic oscillator solution with frictional damping computed using **rk2dyn.m** (*dots*) with adaptive timestep, along with the exact solution (*curve*)

3.6 Dynamic Timestep

For time-dependent problems (ODE initial value problems and parabolic, hyperbolic, and mixed type PDEs), the timestep Δt should always be adjusted dynamically.

For hyperbolic PDEs, explicit methods are recommended, and the timestep is determined dynamically by the CFL condition for stability

$$\Delta t = r \frac{\Delta x}{\max\{|\lambda|\}} \tag{3.33}$$

where the Courant number r satisfies[1] $0 < r \leq 1$ and $\max\{|\lambda|\}$ is the maximum characteristic speed in magnitude on the grid at the beginning of the timestep. Typically r is chosen to be $0.1 \leq r \leq 0.9$.

There are three popular methods for adjusting Δt for ODE initial value problems and parabolic PDEs based on an estimate of the local error: Two computed solutions at t_{n+1} can be compared to estimate e_l, using either the classic Milne strategy below

[1] For some hyperbolic schemes $0 < r \leq \frac{1}{2}$, etc., for stability.

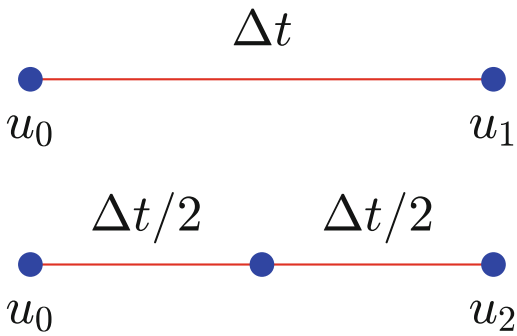

Fig. 3.3 Time levels for dynamic timestep

or using two numerical solutions computed with different methods (e.g., RK4 and RK5); or a divided-difference formula can be used in the TRBDF2 method.

Suppose we have two estimates (Fig. 3.3) of the exact solution $\bar{u}$ at time level $n+1$, based on initial data u_0 at time level n: u_1 calculated with a single step of Δt, and u_2 calculated with two steps of $\Delta t/2$. Then using a technique introduced by Milne, we can estimate the local error $e_l = \bar{u} - u_1 \sim u_2 - u_1$. In fact, for a pth-order method,

$$e_l = \bar{u} - u_1 \approx C\Delta t^{p+1} \tag{3.34}$$

$$u_2 \approx \bar{u} - 2e_l(\Delta t/2) \tag{3.35}$$

$$u_2 - u_1 \approx e_l(\Delta t) - 2e_l(\Delta t/2) \approx \left(1 - \frac{1}{2^p}\right) e_l \tag{3.36}$$

$$e_l \approx \frac{2^p}{2^p - 1}(u_2 - u_1). \tag{3.37}$$

To dynamically adjust Δt, we monitor the estimate of the local error e_l. Define the local ratio r_l, which should be less than or on the order of 1:

$$r_l = \frac{||e_l||}{\epsilon_R||u_n|| + \epsilon_A} \lesssim 1 \tag{3.38}$$

where u_n is the initial data at time level n, ϵ_R specifies the relative error tolerance, and ϵ_A specifies an absolute error tolerance (*effective zero*), so that the denominator of r_l never equals zero. In other words, we are requiring

$$||e_l|| \lesssim \epsilon_R||u_n|| + \epsilon_A. \tag{3.39}$$

Our dynamic timestep algorithm is

- If $r_l \leq 2$, accept the provisional solution $u_{n+1} = u_2$ and set the new $\Delta t \leftarrow \Delta t / r_l^{1/(p+1)}$.
- If $r_l > 2$, redo the timestep with $\Delta t \leftarrow \Delta t / r_l^{1/(p+1)}$ or $\Delta t \leftarrow \Delta t/2$.

The scaling of Δt maintains $||e_l|| \approx \epsilon_R ||u||$, since we are extrapolating that at time level $n+2$

$$r_{l,new} \approx \frac{||e_{l,new}||}{\epsilon_R ||\tilde{u}|| + \epsilon_A} \approx \frac{||e_{l,new}||}{||e_l||} r_l \approx \frac{\Delta t_{new}^{p+1}}{\Delta t^{p+1}} r_l = 1 \tag{3.40}$$

for

$$\Delta t_{new} = \frac{\Delta t}{r_l^{1/(p+1)}} \tag{3.41}$$

where $\tilde{u} \approx u_n \approx u_{n+1}$.

Figure 3.4 illustrates the dynamic timestepping algorithm with RK2/RK2 for the nonlinear van der Pol oscillator described in Sect. 3.7.2. Notice how the timestep opens up when the solution evolves slowly, while it closes down when the solution

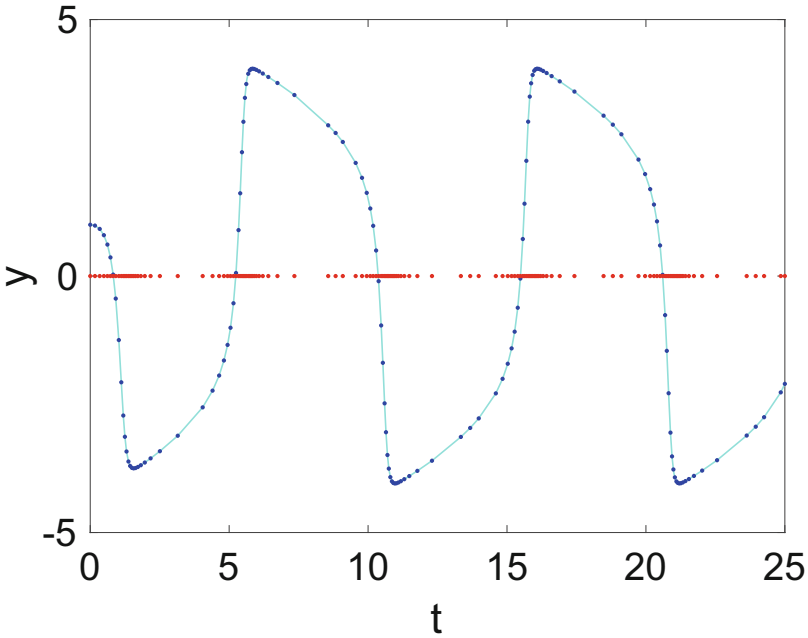

Fig. 3.4 Van der Pol oscillator solution computed using **rk2dyn.m** with adaptive timestep: solution values (*dots* connected by the *curve*), and time levels (*dots* along the time axis). Notice how the timestep opens up in the smooth regions of the solution

Table 3.2 Global relative errors $e_R = |u(t_n) - u_n|/|u(t_n)|$ for van der Pol oscillator simulations using MATLAB's **ode45** RK4/RK5 solver for different values of ϵ_R and $\epsilon_A = \epsilon_M$. Here $u(t_n = 10) = 1.3855149156118856$ to 17 digits from the *Mathematica* ODE solver

ϵ_R	*Steps*	e_R
10^{-3}	265	5.4×10^{-4}
10^{-6}	745	3.7×10^{-6}
10^{-9}	2757	2.6×10^{-9}
10^{-12}	10,845	2.2×10^{-12}

varies rapidly. Table 3.2 demonstrates how the local relative error criterion ϵ_R influences the global relative error e_R for MATLAB's **ode45** solver, where the two different numerical solutions at the new time level are produced with RK4 (u_1) and RK5 (u_2).

The program **rk2dyn.m** solves the damped harmonic oscillator, van der Pol, and Shaw oscillator problems, as well as the Lorenz equations, with a variable Δt (see Sect. 3.7 for a description of the nonlinear dynamical systems). To see the timestep information in **rk2dyn.m**, turn on verbose mode by setting `VERBOSE = 1` at the top of the program.

For **rk2dyn.m**, try

```
rk2dyn([1; -0.05],8*pi,1)   % damped harmonic oscillator
rk2dyn([1; 0],50,2)         % van der Pol oscillator
rk2dyn(1,2,3)               % singularity detection
rk2dyn([0; 1; 0],40,4)      % Lorenz equations
```

where singularity detection is for $y' = y^2$, $y(0) = 1$, $y(t) = 1/(1-t)$.

For the initial Δt_0, recall we want $||e_l|| \approx \epsilon_R ||u||$. A natural timescale for the initial value problem $du/dt = f(u)$ is

$$\bar{t} = \frac{||u_0|| + \frac{\epsilon_A}{\epsilon_R}}{||f_0|| + \frac{\epsilon_A}{\epsilon_R}}. \tag{3.42}$$

Set $\Delta t_0 = \epsilon_R^{1/(1+p)} \bar{t}$ for a pth-order method. Then

$$||e_l|| = C \Delta t_0^{p+1} ||u_0^{(p+1)}|| = C \epsilon_R \bar{t}^{\,p+1} ||u_0^{(p+1)}|| \sim \epsilon_R ||u_0||. \tag{3.43}$$

For implicit methods for parabolic PDEs like linear ($D(u) = const$) and nonlinear diffusion $u_t = (D(u)u_x)_x$, the initial timestep can be taken to be

$$\Delta t_0 = \frac{\sqrt{N}\Delta x^2}{2D_{max}} \tag{3.44}$$

where N is the number of Δx and D_{max} is the maximum value of $D(u)$ on the grid at t_0.

3.7 Deterministic Dynamical Systems

We will always consider *deterministic* dynamical systems $dw/dt = f(w)$ with a continuously differentiable (and therefore Lipschitz continuous) $f(w)$: i.e., the Jacobian $J = \partial f/\partial w$ exists and is continuous. The relevant theorem on existence and uniqueness of solutions is due to Picard and Lindelöf:

Theorem *Picard-Lindelöf Theorem: A unique solution to the initial value problem $dw/dt = f(t, w)$, $w(t_0) = w_0$, exists for $t_0 - \epsilon \leq t \leq t_0 + \epsilon$ for some $\epsilon > 0$ if f is uniformly Lipschitz continuous in w and continuous in t.*

The ODE initial value problem

$$\frac{dy}{dt} = y^{\frac{1}{3}}, \quad y(0) = 0 \tag{3.45}$$

is an example of a nondeterministic problem that has three possible solutions for $t > 0$:

$$y(t) = \begin{cases} 0 \\ \pm\left(\frac{2}{3}t\right)^{\frac{3}{2}} \end{cases} \tag{3.46}$$

This occurs because $y^{\frac{1}{3}}$ is continuous (guarantees existence by the Peano Existence Theorem) but not Lipschitz continuous at $y = 0$ (fails the Picard-Lindelöf uniqueness test).

Poincaré introduced the idea of analyzing the geometry of the phase space of solution trajectories to a deterministic dynamical system. We will illustrate this type of analysis in the context of attractors (limit points, limit cycles, limit tori, and strange attractors) in phase space.

3.7.1 Nonlinear Pendulum

The nonlinear pendulum obeys the initial value problem

$$u'' + \gamma u' + \sin(u) = 0, \quad u(0) = u_0, \quad u'(0) = v_0 \tag{3.47}$$

where $u(t)$ is the angular displacement of the pendulum from the vertical and γ is the coefficient of frictional damping. The modern approach is to rewrite the second-order ODE (3.47) as a system of two first-order equations

$$y' = \begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = f(y) = \begin{bmatrix} y_2 \\ -\sin(y_1) - \gamma y_2 \end{bmatrix}. \tag{3.48}$$

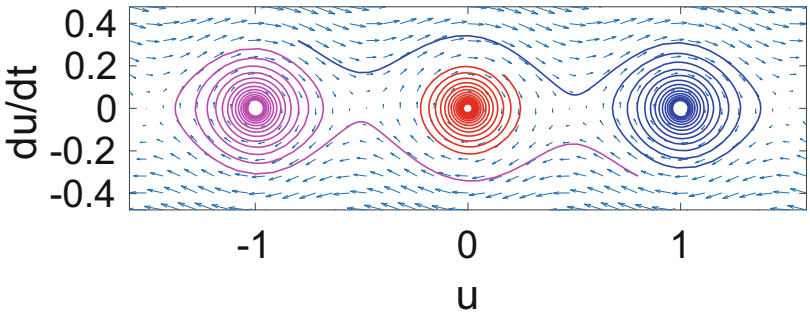

Fig. 3.5 Phase space diagram and direction field for the nonlinear pendulum with frictional damping computed using **ode45**. Note the three limit points in the diagram. u and du/dt have been divided by 2π

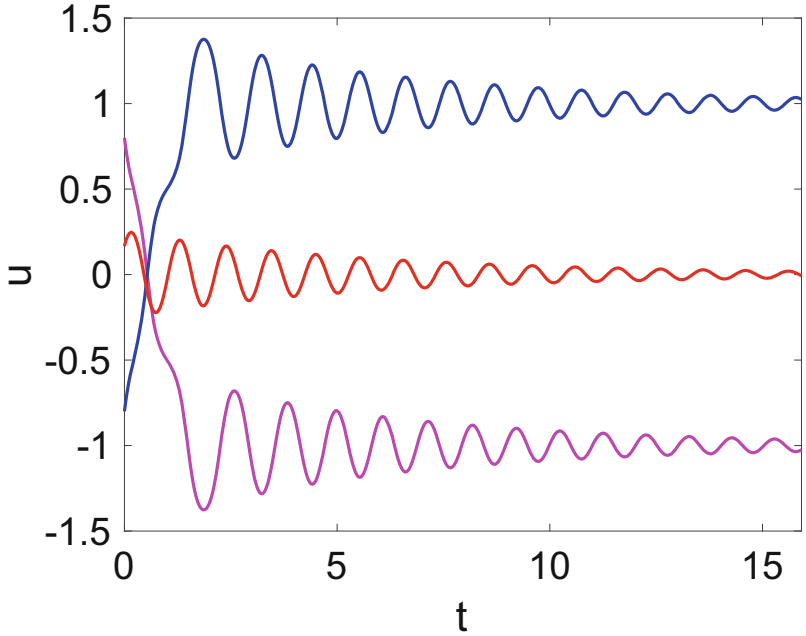

Fig. 3.6 Solutions $u(t)$ for the nonlinear pendulum with frictional damping computed using **ode45**. u and t have been divided by 2π

Three solutions computed with **ode45** are shown in Figs. 3.5 and 3.6.

3.7.2 Van der Pol Oscillator

The van der Pol equations, originally derived for a triode oscillator, are

$$\frac{d^2y}{dt^2} + (y^2 - \epsilon)\frac{dy}{dt} + y = 0, \quad y(0) = 1, \quad \frac{dy}{dt}(0) = 0. \tag{3.49}$$

For $y^2 > \epsilon$, the $(y^2 - \epsilon)y'$ term acts as (nonlinear) friction causing $y \to 0$, while for $y^2 < \epsilon$, the $(y^2 - \epsilon)y'$ term acts as negative friction causing $y \to \pm\infty$. The competition between the two (positive and negative) frictional terms produces a closed loop limit cycle in phase space.

Reexpress the second-order ODE as two first-order ODEs with $u = y$, $v = dy/dt$:

$$\frac{du}{dt} = v \tag{3.50}$$

$$\frac{dv}{dt} = (\epsilon - u^2)v - u \tag{3.51}$$

with initial conditions $u(0) = 1$, $v(0) = 0$. The attractor in phase space exhibiting an asymptotically periodic orbit is shown in Fig. 3.7 for $\epsilon = 4$ (take $t_f = 40$).

3.7.3 Shaw Oscillator

In the van der Pol equations, set $u \to v$ and $v \to -u$, and then add the sinusoidal forcing term to the (wrong!) new dv/dt equation to obtain (up to constants) the Shaw oscillator equations:

$$\frac{du}{dt} = 0.7v + 10u(0.1 - v^2) \tag{3.52}$$

$$\frac{dv}{dt} = -u + 0.25\sin(1.57w) \tag{3.53}$$

$$\frac{dw}{dt} = 1. \tag{3.54}$$

With these parameters, the solution has a strange attractor with fractal dimension $\approx$2.6. For initial conditions, take $u(0) = -0.73$, $v(0) = 0$, $w(0) = 0$ (w is time t), with $t_f = 100$ (see Fig. 3.8). We can make w periodic with period $2\pi/1.57$ by identifying $w + 2m\pi/1.57 \equiv w$, $m = \pm 1, \pm 2, \ldots$ With w periodic, the 3D attractor

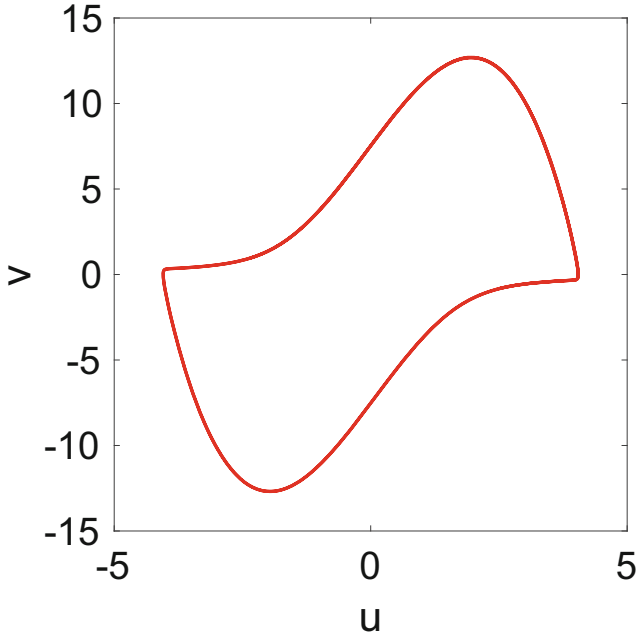

Fig. 3.7 Van der Pol oscillator limit-cycle attractor in the u-v phase space computed with **rk2dyn.m**. The transient solution points have been discarded to obtain the large time attractor

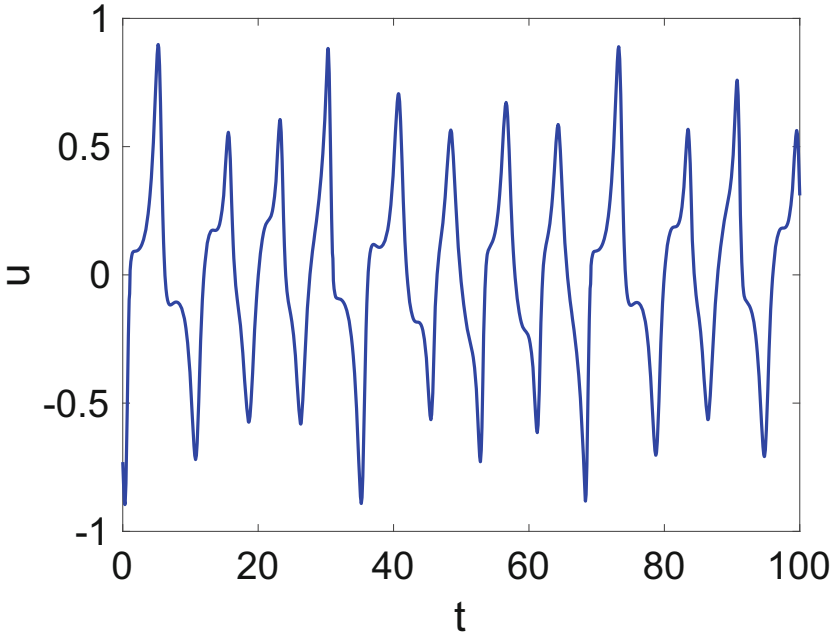

Fig. 3.8 Shaw oscillator solution computed with **ode45**

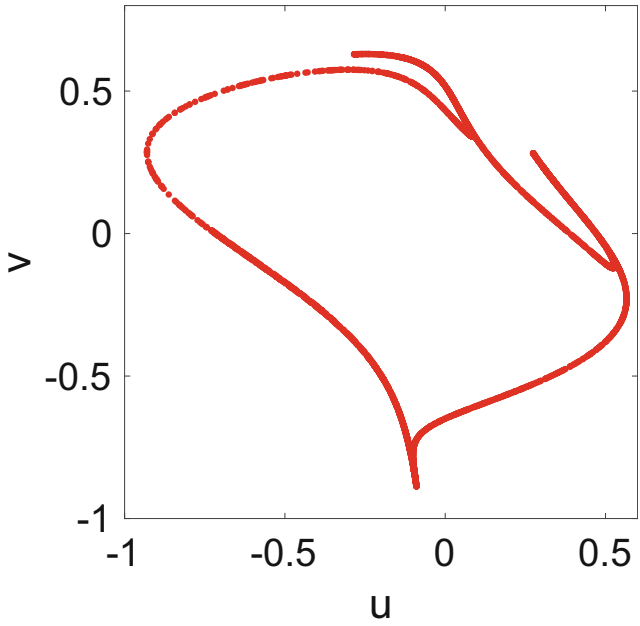

Fig. 3.9 Poincaré slice of Shaw oscillator strange attractor in the u-v phase space computed with **rk2.m**. The transient solution points have been discarded. The cusps are indicative of a strange (fractal dimension) attractor

in the u-v-w phase space is a torus with cusps. The Poincaré slice (Fig. 3.9) of the attractor in the u-v phase space indicates a strange (fractal dimension) attractor.

3.7.4 Lorenz Equations

The Lorenz equations (1963) [28] model vertical counter-rotating vortices in Rayleigh-Bénard convection:

$$\frac{dx}{dt} = \sigma(y - x) \tag{3.55}$$

$$\frac{dy}{dt} = x(r - z) - y \tag{3.56}$$

$$\frac{dz}{dt} = xy - bz. \tag{3.57}$$

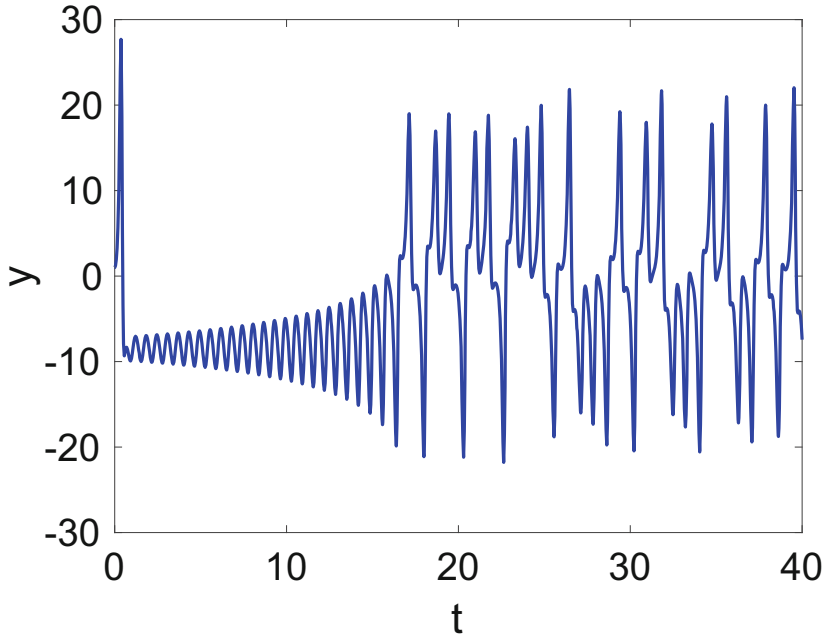

Fig. 3.10 Lorenz equations solution $y(t)$ computed with **ode45**

For a Rayleigh-Bénard vortex, $x(t) \sim$ the angular velocity, $y(t \to \infty) \sim T$ at the middle right edge, and $z(t \to \infty) \sim T$ at the bottom. Here σ is the Prandtl number[2] (10 is appropriate for cold water; too high for air), r is the Rayleigh number,[3] and b is the aspect ratio of the vortex cell.

For $\sigma = 10$, $r = 28$, and $b = 8/3$, the solution (Fig. 3.10) has a strange attractor (Fig. 3.11) with fractal dimension $\approx$2.06. For initial conditions, take $x(0) = 0$, $y(0) = 1$, and $z(0) = 0$, with $t_f = 30$.

3.7.5 *Stability of Equilibria*

The equilibria w_{eq} of an initial value problem system $dw/dt = f(w)$ are given by $f(w_{eq}) = 0$. To analyze the stability of the equilibria, make a small perturbation: $w(t) = w_{eq} + \delta w(t)$. Linearizing with respect to δw, we obtain

[2] Dimensionless ratio ν/κ of momentum diffusivity (kinematic viscosity) to thermal diffusivity.

[3] Dimensionless number describing heat flow—if the Rayleigh number is below (above) a critical value, heat transfer is dominated by conduction (convection).

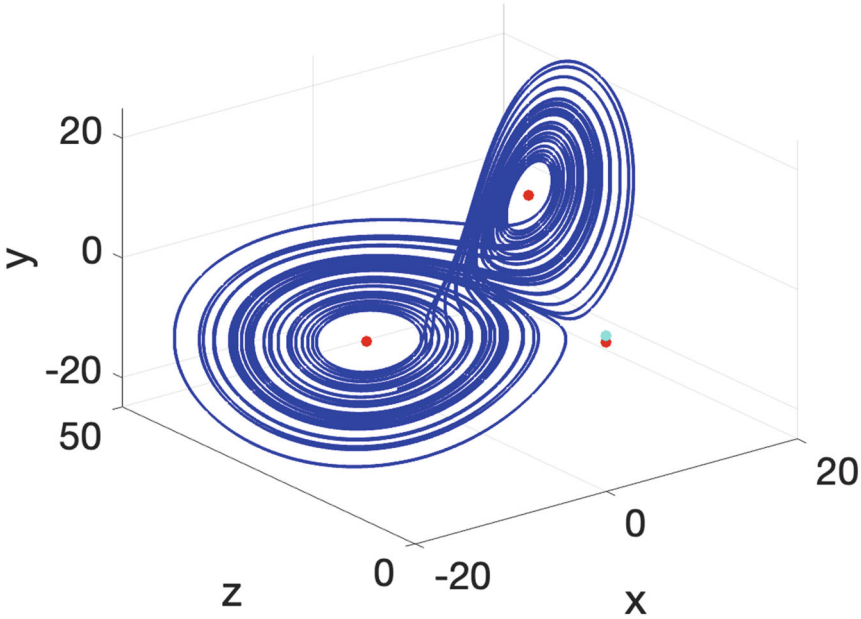

Fig. 3.11 Lorenz equations strange attractor in the x-y-z phase space computed with **ode45**. The transient solution points have been discarded. The *dark dots* are the three equilibria, and the *light dot* represents the initial conditions

$$\frac{d\delta w}{dt} = f(w_{eq} + \delta w) \approx J\,\delta w, \quad J = \left.\frac{\partial f}{\partial w}\right|_{w_{eq}}. \tag{3.58}$$

Then set $\delta w(t) = e^{\lambda t}\widetilde{w}$ where $\widetilde{w}$ is a constant vector. We obtain an eigenvalue problem

$$\lambda \delta w = J\,\delta w. \tag{3.59}$$

If $\Re\{\lambda_j\} < 0$ for all j, then the equilibrium is stable. If $\Re\{\lambda_j\} > 0$ for any j, then the equilibrium is unstable.

Write the Lorenz equations (3.55)–(3.57) as

$$\frac{dw}{dt} = \frac{d}{dt}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = f(w) = \begin{bmatrix} \sigma(y-x) \\ x(r-z)-y \\ xy - bz \end{bmatrix}. \tag{3.60}$$

There are three equilibria $f(w_{eq}) = 0$:

$$(x, y, z)_{eq} = (\eta, \eta, r-1), \quad (-\eta, -\eta, r-1), \quad (0, 0, 0) \tag{3.61}$$

where $\eta = \sqrt{b(r-1)}$. The eigenvalues λ_j ($j = 1, 2, 3$) of the Jacobian

$$J = \frac{\partial f}{\partial w} = \begin{bmatrix} -\sigma & \sigma & 0 \\ r - z & -1 & -x \\ y & x & -b \end{bmatrix} \tag{3.62}$$

evaluated at $J(w_{eq})$ determine the stability of the equilibria. All three equilibria are unstable for the canonical values $\sigma = 10$, $r = 28$, and $b = 8/3$, with the centers of the "butterfly wings" $(\eta, \eta, r-1)$ and $(-\eta, -\eta, r-1)$ slightly unstable and the origin $(0, 0, 0)$ extremely unstable (run **lorenzeigs.m** for an analysis).

3.8 Backward Euler and Newton's Method

When we apply an implicit method to a nonlinear initial value problem $du/dt = f(u)$, the nonlinear discretized equation for u_{n+1} is best solved using Newton's method.

We start with the simple example of applying backward Euler and Newton's method to solving a nonlinear initial value problem; then in the next section, we extend the methodology to TRBDF2.

To integrate $du/dt = f(u)$ from $t = t_n$ to $t_{n+1} = t_n + \Delta t$, the backward Euler formula is

$$u_{n+1} = u_n + \Delta t\, f(u_{n+1}). \tag{3.63}$$

We linearize $f(u_{n+1})$ in (3.63) iteratively by setting the new iterative solution to

$$u_{n+1}^{(k+1)} = u_{n+1}^{(k)} + \Delta u^{(k)}, \quad u_{n+1}^{(0)} = u_n, \quad k = 0, 1, 2, \ldots \tag{3.64}$$

and approximating

$$f\left(u_{n+1}^{(k+1)}\right) = f\left(u_{n+1}^{(k)}\right) + \left(\frac{\partial f}{\partial u}\right)_{n+1}^{(k)} \Delta u^{(k)} \tag{3.65}$$

where $k = 0, 1, 2, \ldots$ labels the Newton iterations. Then, we iterate until the Newton method converges.

The Newton equation for backward Euler is

$$\left[I - \Delta t \left(\frac{\partial f}{\partial u}\right)_{n+1}^{(k)}\right] \Delta u^{(k)} = -u_{n+1}^{(k)} + u_n + \Delta t f\left(u_{n+1}^{(k)}\right) \tag{3.66}$$

or

$$\left(\frac{\partial R_{\mathrm{BE}}}{\partial u_{n+1}}\right)_{n+1}^{(k)} \Delta u^{(k)} = -R_{\mathrm{BE}} \tag{3.67}$$

where $R_{\mathrm{BE}} = u_{n+1}^{(k)} - u_n - \Delta t\, f\left(u_{n+1}^{(k)}\right)$ is the residual for (3.63). Note that (3.67) has the symbolic Newton form $R'\,\Delta u = -R$.

The Jacobian can be evaluated by finite differences in the following way if an analytical Jacobian is not available:

$$J_{ij} = \frac{\partial f_i}{\partial u_j} \approx \frac{\Delta f_i}{\Delta u_j} = \frac{f_i(u+\Delta u) - f_i(u)}{\Delta u_j} \approx \frac{\sum_k \frac{\partial f_i}{\partial u_k}\Delta u_k}{\Delta u_j} = \frac{\frac{\partial f_i}{\partial u_j}\Delta u_j}{\Delta u_j} = \frac{\partial f_i}{\partial u_j} \tag{3.68}$$

by setting

$$\Delta u_k = \begin{cases} \Delta u_j & \text{if } k = j \\ 0 & \text{if } k \neq j \end{cases} \tag{3.69}$$

where $i, j, k = 1, 2, \ldots, m$ for m-dimensional u and f and where $\Delta u_j = 10^{-9}$, for example.

3.9 TRBDF2 Method

For *stiff* ODE or PDE initial value problems, TRBDF2 [1] is the method of choice, since it is L-stable and second-order accurate.

Stiffness is a subtle concept. A general, broad formulation is that an initial value problem is stiff if it has two or more widely disparate solution timescales. A more precise, narrower definition is given by Moler (Sect. 7.9 of [29]):

> A problem is stiff if the solution being sought varies slowly, but there are nearby solutions that vary rapidly, so the numerical method must take small steps to obtain satisfactory results.

The best way to understand stiffness is through two examples: (i) the L-stable vs. only A-stable computed solutions of the nonlinear diffusion PDE (Fig. 2.3) and (ii) the flame propagation ODE in Problem 3.16.

To integrate $du/dt = f(u)$ from $t = t_n$ to $t_{n+1} = t_n + \Delta t$ using the TRBDF2 method (Fig. 3.12), first apply the trapezoidal rule (TR) to advance the solution from t_n to $t_{n+\gamma} = t_n + \gamma\,\Delta t$:

$$u_{n+\gamma} - \gamma\frac{\Delta t}{2} f_{n+\gamma} = u_n + \gamma\frac{\Delta t}{2} f_n \tag{3.70}$$

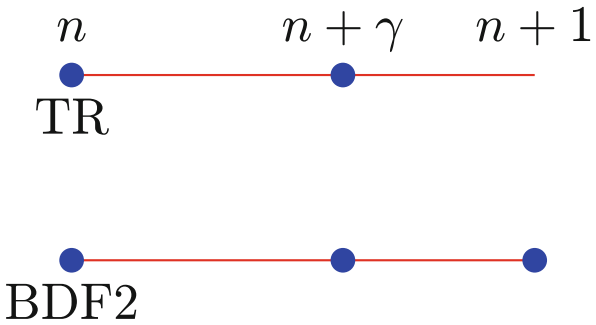

Fig. 3.12 Time levels for TRBDF2

and then use the second-order backward differentiation formula (BDF2)[4] to advance the solution to t_{n+1}:

$$u_{n+1} - \frac{1-\gamma}{2-\gamma}\Delta t\, f_{n+1} = \frac{1}{\gamma(2-\gamma)}u_{n+\gamma} - \frac{(1-\gamma)^2}{\gamma(2-\gamma)}u_n. \tag{3.71}$$

This composite one-step method is second-order accurate and L-stable.

We linearize f_{n+1} in (3.71) (and similarly $f_{n+\gamma}$ in (3.70)) iteratively by setting the new iterative solution to

$$u_{n+1}^{(k+1)} = u_{n+1}^{(k)} + \Delta u^{(k)}, \quad u_{n+1}^{(0)} = u_{n+\gamma} \tag{3.72}$$

and approximating

$$f_{n+1}^{(k+1)} = f_{n+1}^{(k)} + \left(\frac{\partial f}{\partial u}\right)_{n+1}^{(k)} \Delta u^{(k)} \tag{3.73}$$

where $k = 0, 1, 2, \ldots$ labels the Newton iterations. At each TR or BDF2 partial step, we iterate until the Newton method converges.

The Newton equation for the TR partial step is

$$\left[I - \gamma\frac{\Delta t}{2}\left(\frac{\partial f}{\partial u}\right)_{n+\gamma}^{(k)}\right]\Delta u^{(k)} = -\left(u_{n+\gamma}^{(k)} - u_n\right) + \gamma\frac{\Delta t}{2}\left(f_{n+\gamma}^{(k)} + f_n\right) \equiv -R_{\mathrm{TR}} \tag{3.74}$$

where R_{TR} is the residual for (3.70).

The Newton equation for the BDF2 partial step is

[4] For BDF2, $e_l \approx -\frac{(1-\gamma)^2}{6(2-\gamma)}\Delta t^3 u'''$.

$$\left[I - \frac{1-\gamma}{2-\gamma}\Delta t \left(\frac{\partial f}{\partial u}\right)^{(k)}_{n+1} \right] \Delta u^{(k)}$$

$$= -\left(u^{(k)}_{n+1} - \frac{1}{\gamma(2-\gamma)} u_{n+\gamma} + \frac{(1-\gamma)^2}{\gamma(2-\gamma)} u_n \right) + \frac{1-\gamma}{2-\gamma}\Delta t f^{(k)}_{n+1} \equiv -R_{\mathrm{BDF2}} \tag{3.75}$$

where R_{BDF2} is the residual for (3.71).

The timestep size Δt is adjusted dynamically within a window $[\Delta t_{\min}, \Delta t_{\max}]$ by monitoring a *divided-difference* estimate of the local error e_l:

$$e_l = k_\gamma \Delta t^3 u''' \approx 2k_\gamma \Delta t \left(\frac{1}{\gamma} f_n - \frac{1}{\gamma(1-\gamma)} f_{n+\gamma} + \frac{1}{1-\gamma} f_{n+1} \right) \tag{3.76}$$

where

$$k_\gamma = \frac{-3\gamma^2 + 4\gamma - 2}{12(2-\gamma)}. \tag{3.77}$$

The three values of f employed in (3.76) have already been calculated in the most recent TRBDF2 timestep. The $||e_l||$ is minimized for $\gamma = 2 - \sqrt{2} \approx 0.59$.

Recall from (2.24) that a time integration method for $du/dt = au$, $\Re\{a\} < 0$, is *A-stable* if

$$\frac{||u_{n+1}||}{||u_n||} = ||G|| \leq 1 \tag{3.78}$$

for all $\Delta t > 0$; from (2.25), the time integration method is *L-stable* if it is A-stable and

$$\lim_{\Delta t \to \infty} \frac{||u_{n+1}||}{||u_n||} = \lim_{\Delta t \to \infty} ||G|| = 0. \tag{3.79}$$

For the heat/diffusion equation, TR is A-stable but not L-stable; backward Euler (first- and second-order) and TRBDF2 are L-stable.

The growth factor G for TRBDF2 for $du/dt = -\alpha u$, $\alpha > 0$, is given by ($\Delta = \gamma\alpha\Delta t > 0$):

$$G(\Delta t, \gamma) = \frac{\frac{1-\frac{\gamma\alpha\Delta t}{2}}{1+\frac{\gamma\alpha\Delta t}{2}} - (1-\gamma)^2}{\gamma(2-\gamma) + \gamma(1-\gamma)\alpha\Delta t} = \frac{\frac{2-\Delta}{2+\Delta} - (1-\gamma)^2}{\gamma(2-\gamma) + (1-\gamma)\Delta}. \tag{3.80}$$

For $\gamma = 1$, the BDF2 step disappears, and $G \to G_{TR}$; similarly, for $\gamma = 0$, the TR step disappears, the BDF2 step $\to$ TR, and $G \to G_{TR}$. In both cases, $k_\gamma \to -\frac{1}{12}$

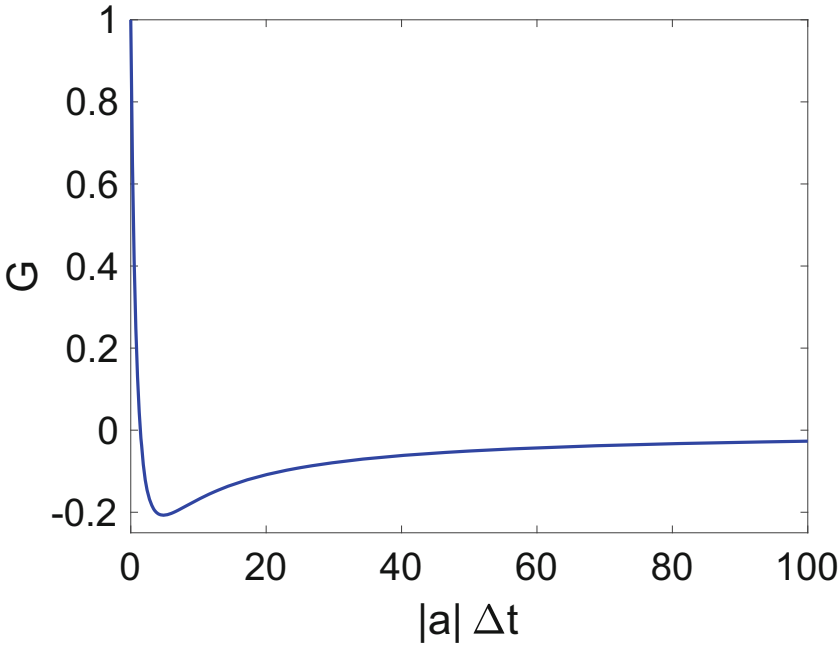

Fig. 3.13 TRBDF2 growth factor G as a function of $|a|\Delta t$ for $\gamma = 2 - \sqrt{2}$

(the TR value). Figure 3.13 illustrates G for $\gamma = 2 - \sqrt{2}$. Problem 3.18 derives G and proves that TRBDF2 is A-stable and L-stable.

Figure 3.1 for TRBDF2 with $\gamma = 2-\sqrt{2}$ demonstrates graphically that TRBDF2 is A-stable.

3.10 Equivalence Theorem for ODE IVPs

Theorem *Equivalence Theorem: For* consistent *numerical approximations,* stability *and* convergence *are equivalent.*

Lax proved the Equivalence Theorem for ODE initial value problems. The theorem applies as well to ODE boundary value problems, approximations to functions and integrals, and PDE initial value problems. Here we will prove the theorem for ODE initial value problems; the proof (due to Richtmyer) for PDE initial value problems is more difficult, since it involves functional analysis (see Sect. 5.3 on the Equivalence Theorem for PDEs).

We give two proofs, following Strang [35].

Approximate $Lu = f$ by $L_\Delta u_\Delta = f_\Delta$. *Consistency* implies $f_\Delta \to f$ and $L_\Delta u \to Lu$ as $n \to \infty$, $\Delta t \to 0$, with $n\Delta t = t$ fixed. *Stability* implies L_Δ^{-1}

remains uniformly bounded. Then as $n \to \infty$, $\Delta t \to 0$, with $n\Delta t = t$ fixed, u_Δ *converges* to u:

$$u - u_\Delta = L_\Delta^{-1}(L_\Delta u - Lu) + L_\Delta^{-1}(f - f_\Delta) \to 0. \tag{3.81}$$

Let's look at the details for the initial value problem $du/dt = au$, $u(0) = u_0$. The exact solution is $u(t) = e^{at}u_0 = H^n u_0$, where $H = e^{a\Delta t}$ is the exact growth factor. The approximate solution is $u_n = G^n u_0$. *Stability* is equivalent to

$$|G^n| \le e^{Kn\Delta t} = e^{Kt} \tag{3.82}$$

where K is a nonnegative constant independent of n. For $a > 0$, choose $K \ge a > 0$; for $a < 0$ and A-stability, set $K = 0$ with $e^{Kt} = 1$. Then both $|G|$ and $|H|$ are $\le e^{K\Delta t}$. *Consistency* implies

$$|G - H| \le C\Delta t^{p+1}, \quad p > 0. \tag{3.83}$$

Convergence is equivalent to

$$G^n u_0 - H^n u_0 \to 0 \tag{3.84}$$

as $n \to \infty$, $\Delta t \to 0$, with $n\Delta t = t$ fixed.

To show *stability* implies *convergence*, we use a telescoping identity:

$$\begin{aligned} G^n - H^n &= G^n - G^{n-1}H + G^{n-1}H - G^{n-2}H^2 + \ldots + GH^{n-1} - H^n \\ &= G^{n-1}(G-H) + G^{n-2}(G-H)H + \ldots + (G-H)H^{n-1}. \end{aligned} \tag{3.85}$$

There is a factor of $G-H$ in every term, with $|G-H| \le C\Delta t^{p+1}$ by consistency. There is a nonnegative integer power of G in every term, which is bounded by stability. And there is a nonnegative integer power of H in every term, which is bounded, since the continuous problem is well-posed. There are $n = t/\Delta t$ terms in (3.85). Therefore as $n \to \infty$, $\Delta t \to 0$, with $n\Delta t = t$ fixed, we have convergence:

$$|G^n - H^n| \le \frac{t}{\Delta t} e^{Kt} C\Delta t^{p+1} = O(\Delta t^p) \to 0. \tag{3.86}$$

Each step was necessary—and all steps together are sufficient—for convergence, so we have proved *stability* is equivalent to *convergence* for *consistent* numerical schemes.

Numerical error accumulates in a difference equation exactly via the telescoping series (3.85):

$$G^n - H^n = \sum_{j=1}^{n} G^{n-j}(G-H)H^{j-1}. \tag{3.87}$$

H^{j-1} propagates the exact solution to time level $j-1$; $(G-H)$ is the local error going from time level $j-1$ to j; and G^{n-j} propagates this error forward with the difference method to time level n.

Problems

3.1 For $du/dt = f(u)$, prove the backward Euler method is first-order accurate, using the definition of the local truncation error LTE = $\Delta t\,\tau$. Be sure to calculate the constant multiplying Δt in the global error τ. *Hint:* Replace $f(u_{n+1}) = u'(t+\Delta t)$.

3.2 Prove backward Euler is A-stable and L-stable for the model problem $du/dt = au$ with $a < 0$.

3.3 The *modified midpoint rule (MMR)* for $du/dt = f(u)$ is

$$u_{n+1} = u_n + \Delta t\, f\left(\frac{1}{2}(u_n + u_{n+1})\right).$$

Note the subtle difference from the TR method (3.12). For the linear initial value problem $du/dt = au$, show MMR is second-order accurate and A-stable but not L-stable.

3.4

(a) Show the Adams-Bashforth-Moulton predictor-corrector method (3.19) and (3.20) is second-order accurate.
(b) Verify the method is stable for $u' = au$, $a < 0$, if $\Delta t \le 2/|a|$.

3.5

(a) Show the fourth-order Adams-Bashforth-Moulton predictor-corrector formulas (3.21) and (3.22) are correct by fitting a cubic to the four most recent points in the derivative picture, extrapolating for (3.21), and then integrating under the cubic polynomials.
(b) Also find the stability restriction on Δt for $u' = au$, $a < 0$.

3.6 For $du/dt = au$, show RK2 is (a) second-order accurate and (b) conditionally stable. *Hints:* For (a), show the growth factor $G = 1 + a\Delta t + \frac{1}{2}a^2\Delta t^2 \approx e^{a\Delta t}$ to second order in Δt; in (b), take $a < 0$, and require $-1 \le G \le 1$.

3.7 Show RK4 is fourth-order accurate for $du/dt = au$ by showing the growth factor $G = e^{a\Delta t}$ through fourth order in Δt. Also find the stability condition on Δt.

3.8 Run the program **ivp.m** that solves the initial value problem $dy/dt = -y$, $y(0) = 1$ using backward Euler and TR. First try `ivp(10,2,1)` to see a typical case with 10 steps, $t_f = 2$, and $\Delta t = 0.2$.

(a) What goes wrong at $\Delta t = 2$? Try `ivp(10,20,1)` with ten steps, $t_f = 20$, $\Delta t = 2$. *Hint:* Look at G_{BE} and G_{TR}. For $y' = ay$,

$$G_{BE} = \frac{1}{1 - a\Delta t}, \quad G_{TR} = \frac{1 + \frac{1}{2}a\Delta t}{1 - \frac{1}{2}a\Delta t}.$$

(b) What do the results for $\Delta t = 2.1$ illustrate? Try `ivp(10,21,1)` with ten steps, $t_f = 21$, $\Delta t = 2.1$. *Hint:* Look at G_{BE} and G_{TR}.

3.9 For the van der Pol oscillator in **rk2dyn.m** with $\epsilon = 4$, explore various initial conditions: for example, $w_0 = [1; 0]$, $w_0 = [1; -5]$, $w_0 = [-2; 5]$, etc. What can you conclude about the various initial conditions and the long time attractor? Discuss why this happens based on the competition (3.49) between positive friction (when $y^2 > \epsilon$) and negative friction (when $y^2 < \epsilon$).

3.10 Verify that our **rk2dyn.m** code for the nonlinear van der Pol oscillator (take $\epsilon = 4$) converges under mesh refinement by comparing and contrasting simulations with $\epsilon_R = 10^{-2}$, 10^{-3}, and 10^{-6} at $t = 50$ (keep $\epsilon_A = 10^{-9}$). Use `RK2dyn([1;0],50,2)`.

3.11 Investigate the limits of predictability for the Lorenz equations using **lorenz2.m** by comparing the graphs of $y(t)$ with initial data $x(0) = 0$, $y(0) = 1$, $z(0) = 0$, and $y_\delta(t)$ with initial data $x(0) = 0$, $y(0) = 1 + \delta$, $z(0) = 0$, where $\delta = 10^{-6}$. Set $\sigma = 10$, $r = 28$, $b = 8/3$, $t_f = 40$, $\epsilon_R = 10^{-12}$, and $\epsilon_A = 10^{-15}$. Describe your results.

3.12 Find the largest δ to the nearest power of 10 (e.g., 10^{-15}), so that the plots of $y(t)$ and $y_\delta(t)$ agree to within a line width out to $t_f = 40$. Set $\sigma = 10$, $r = 28$, $b = 8/3$, $t_f = 40$, $\epsilon_R = 10^{-12}$, and $\epsilon_A = 10^{-15}$. Discuss your results.

3.13 Investigate the limits of predictability for the Lorenz equations using **lorenz3.m** by comparing the graphs of $y(t)$ computed with **ode45**, **ode23tb** (with analytical Jacobian), and **ode23tb** (with finite difference Jacobian). Set $\sigma = 10$, $r = 28$, $b = 8/3$, $t_f = 40$, $\epsilon_R = 10^{-12}$, and $\epsilon_A = 10^{-15}$. Take the initial data $x(0) = 0$, $y(0) = 1$, $z(0) = 0$. Describe your results.

3.14 Investigate fixed point and periodic orbits for the Lorenz equations by varying r with $\sigma = 10$ and $b = 8/3$. Try $r = 10$ (fixed point), $r = 350$ (simplest periodic orbit), and $r = 99.65$, 100.5, and 160 (complicated periodic orbits).

3.15 Show the Jacobian for the Lorenz equations is given by (3.62).

3.16 Apply **ode45** (explicit RK4/RK5) and **ode23tb** (implicit TRBDF2) to the stiff initial value problem (Moler [29], Sect. 7.9) governing the radius $y(t)$ of a spherical ball of flame:

$$y' = y^2 - y^3, \quad y(0) = \delta.$$

The flame propagation equation says that the radius of the ball of flame increases in proportion to its surface area (the $+y^2$ term) and decreases in proportion to its volume (the $-y^3$ term). Try

```
delta = 0.00001;
F = @(t,y) y^2 - y^3;
ode45(F,[0 1.2/delta],delta);
```

Then try the stiff solver

```
ode23tb(F,[0 1.2/delta],delta);
```

In each case, zoom in on $t = 1/\delta$ to see the local oscillations with the nonstiff solver **ode45**, and their cure with the stiff solver **ode23tb**.

3.17 For the initial value problem $du/dt = f(u)$, derive the leading local error term (including the constant) for TRBDF2 using the definition of the LTE:

$$e_l \approx \text{LTE} \approx k_\gamma \Delta t^3 u''', \quad k_\gamma = \frac{-3\gamma^2 + 4\gamma - 2}{12(2-\gamma)}.$$

Hint: Set $u_{n+\gamma} = u(t_{n+\gamma}) - e_l^{TR}$.

3.18 Derive the growth factor G for TRBDF2 for $du/dt = -\alpha u$, $\alpha > 0$, and show the method is A-stable and L-stable for $0 < \gamma < 1$. *Hints:* (i) First derive the growth factor (3.80). (ii) Then write the A-stability conditions as

$$-1 \le G = \frac{\frac{2-\Delta}{2+\Delta} - \theta_1^2}{\theta_2 + \theta_1 \Delta} \le 1$$

where $\Delta = \gamma\alpha\Delta t > 0$ and

$$0 < \theta_1 = 1 - \gamma < 1, \quad 0 < \theta_2 = \gamma(2-\gamma) < 1.$$

Note that

$$\theta_1^2 + \theta_2 = 1, \quad \theta_1 + \theta_2 - 1 = \gamma(1-\gamma) > 0.$$

(iii) Finally prove $-1 \le G \le 1$.

3.19

(a) Verify the TRBDF2 divided-difference estimate (3.76) of the local error e_l,
(b) Show $|k_\gamma|$ (and thus $||e_l||$) is minimized for $0 < \gamma = 2 - \sqrt{2} < 1$.

3.20 For $du/dt = f(u)$, show

$$||e_l|| = ||u(t_{n+1}) - u_{n+1}|| \approx \frac{3\gamma^2 - 4\gamma + 2}{3\gamma(1-\gamma)^2} \, ||u_1 - u_2||$$

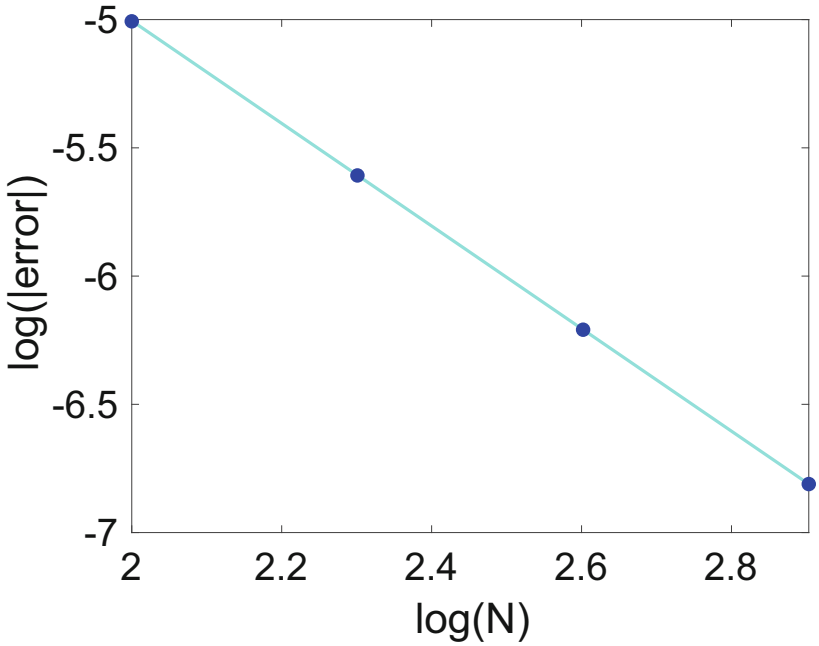

Fig. 3.14 Numerical errors $\log_{10}(||y_{exact} - y_{comp}||_1)$ vs. $\log_{10}(N)$ for TR using **ivp.m** for $dy/dt = -y$, $y(0) = 1$, with $N = 100$, 200, 400, and $800\Delta t$ at $t = 2$. The *straight line* is the theoretical value with log-log slope of -2

for the TR/TR version of calculating the local error for TRBDF2, where $u_1 \equiv u_{n+1} = u_{TRBDF2}$ and $u_2 \equiv u_{TR/TR}$ (see [23]).

3.21 In the TRBDF2 notes, verify (3.74) and (3.75) from the general Newton's formula for $du/dt = f(u)$

$$\frac{\partial R}{\partial u}\Delta u = -R$$

where R is the residual of the method (TR or BDF2).

3.22 Verify that our implementations of forward and backward Euler, TR, and TRBDF2 achieve the correct orders of accuracy Δt^p by comparing computed and exact solutions at $t = 2$ for the model problem $y' = -y$, $y(0) = 1$ on various temporal grids (Fig. 3.14). *Note:* The same exercise can be applied to the (damped) harmonic oscillator in **rk2.m** where an exact solution is available and to the nonlinear pendulum and van der Pol oscillator by letting a fine mesh solution stand in for the exact solution.

3.23 *Project:* Upgrade our RK2/RK2 solver to RK4/RK4, and test on the various problem types: damped harmonic oscillator, nonlinear pendulum, van der Pol oscillator, Shaw oscillator, Lorenz equations, and singularity detection.